

PCB Calculation

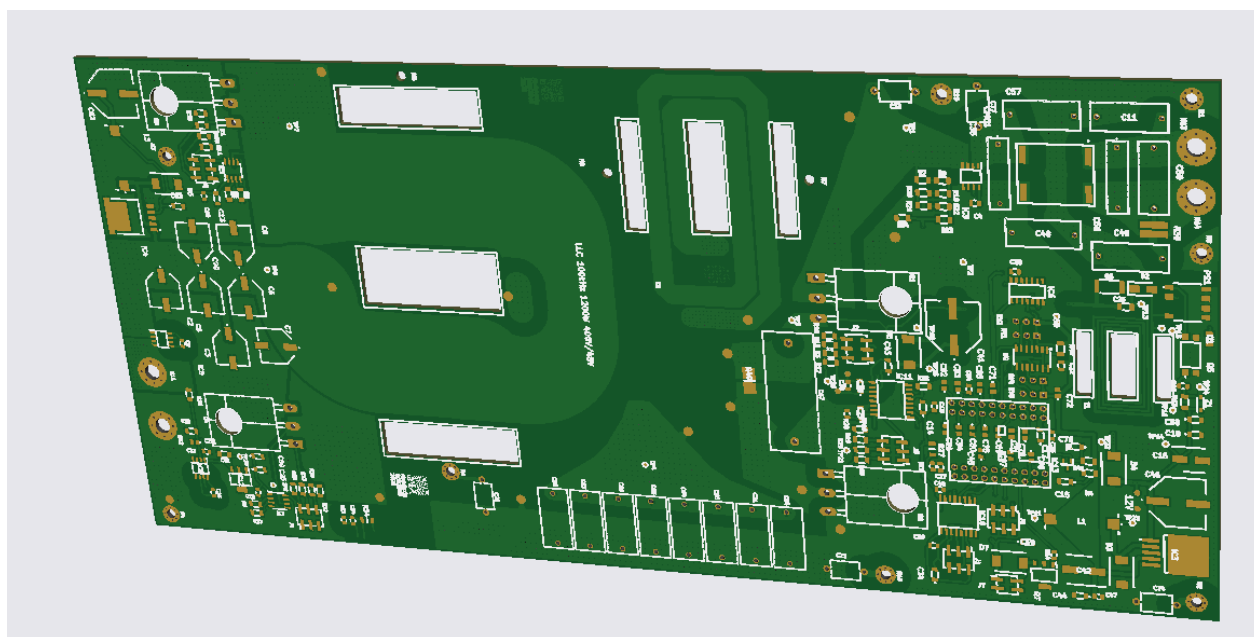


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PCB allowed current

Formula

$$I = K \cdot (\Delta_T)^{0.44} \cdot (A)^{0.725}$$

Where :

I (Current (A))

A (Cross section in sq. mils)

Δ_T (Temperature rise in °C)

$K = 0.048$ (Outer layer)

$K = 0.240$ (Inner layer)

See PC-2221A(L), page 50

$$I = k\Delta T^{0.44} A^{0.725}$$

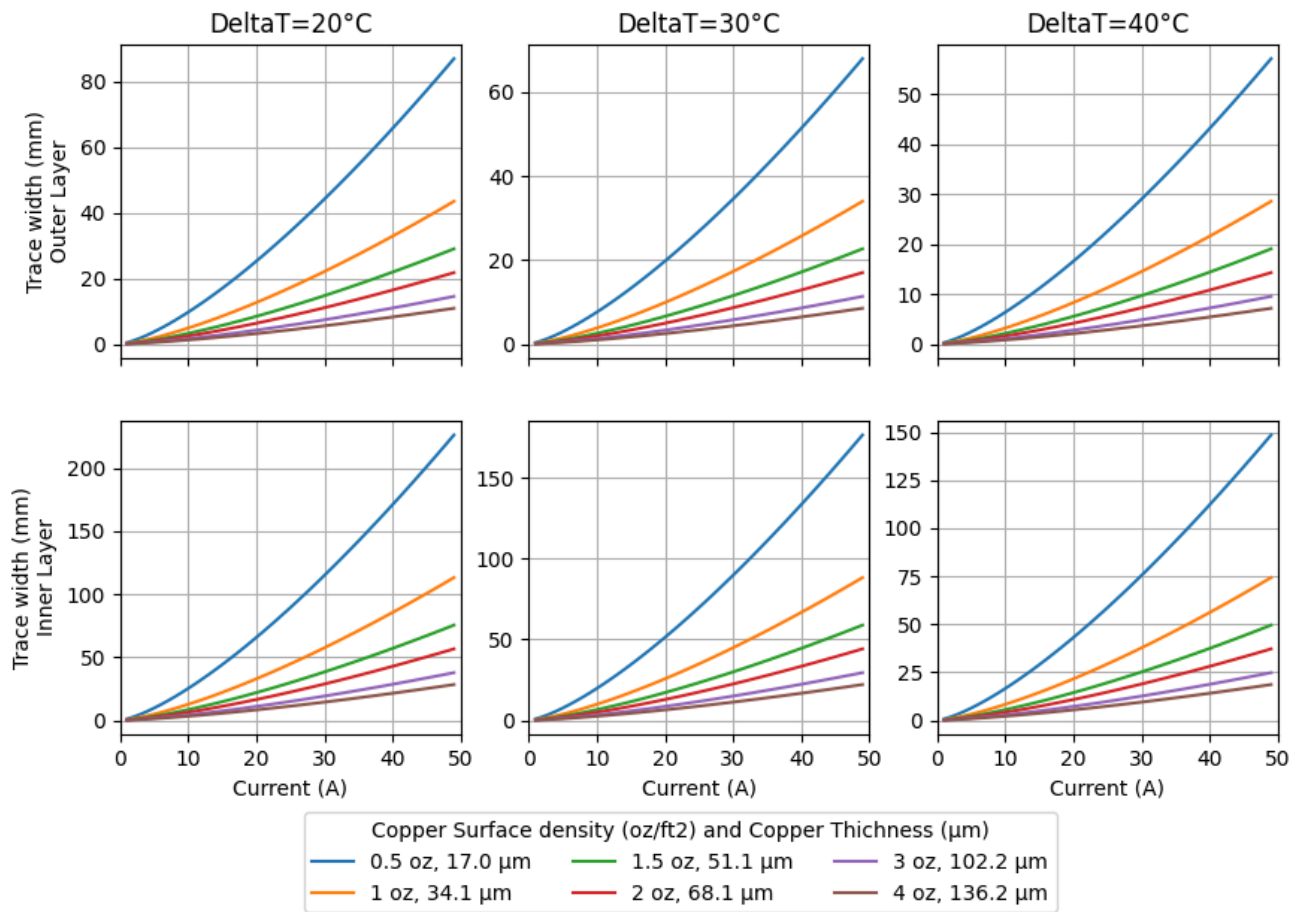
Where I = current in amperes, A = cross section in sq. mils,
and ΔT = temperature rise in °C and k is a constant such
that:

k = 0.048 for outer layers
k = 0.024 for inner layers

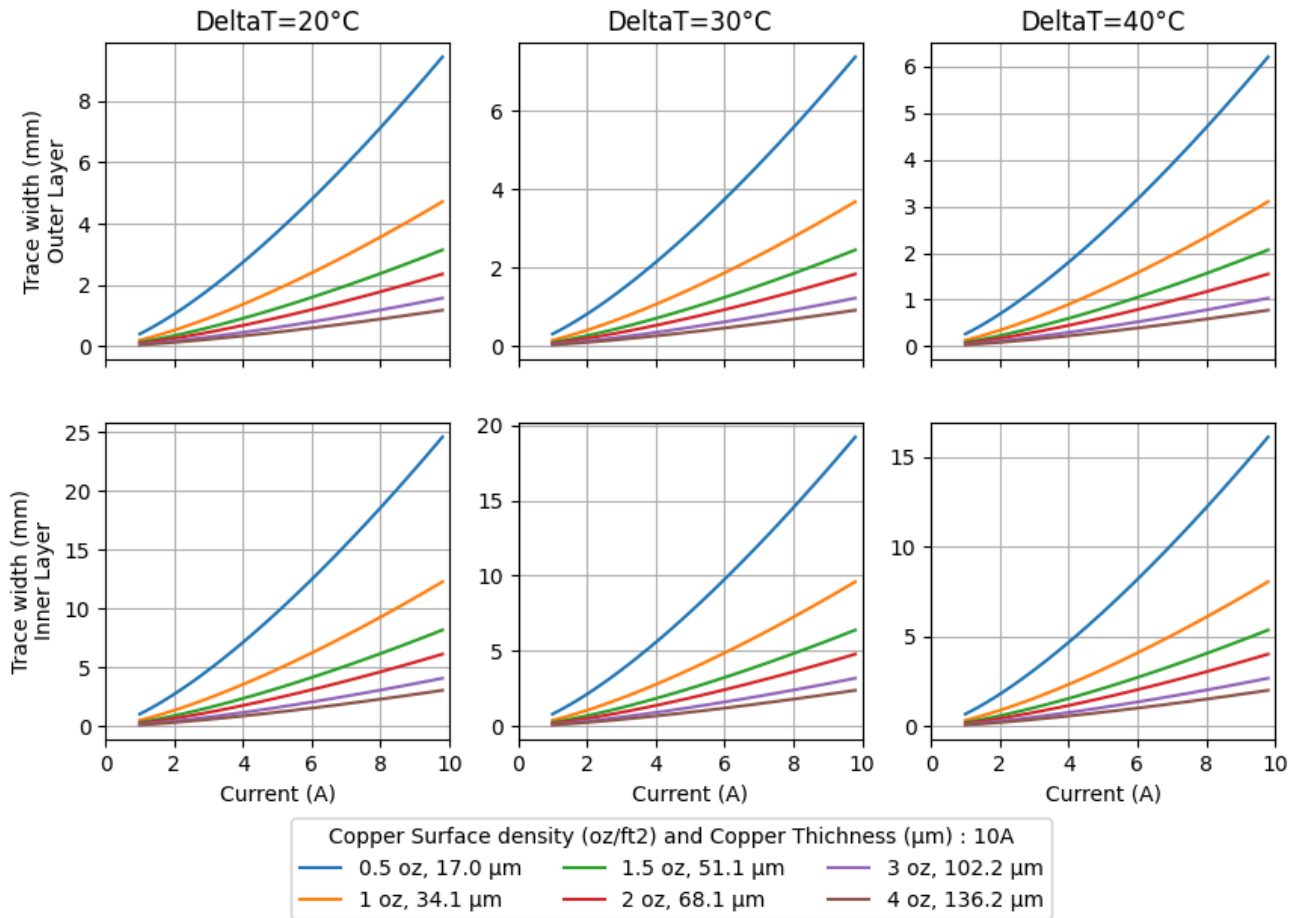
Figure 1

Below an application of this formula for outer / inner layers

Relationship Between PCB Layer Type, Current, Copper Weight, and Minimum Trace Width: 50A



Relationship Between PCB Layer Type, Current, Copper Weight, and Minimum Trace Width: 10A



Application to the actual design

Quick reminder of the nominal value of the DC-DC.

P_{nom} 1200
 Primary: $I_{\text{in_dc}}$ 4.286
 $V_{\text{out_nom}}$ 48
 $V_{\text{in_nom}}$ 380
 Primary: $I_{\text{r_rms}}$ 10.354
 Secondary: $I_{\text{o_dc}}$ 25.0
 Secondary: $I_{\text{os_rms}}$ 30.545

We will use a FR-4 135°C, we will suppose that we have 30°C as ΔT

Layer position = outer , copper = 0.5 oz (Thickness = 17 μ m)
 Primary: Iin_dc 4.286 A => Minimum trace width = 2.36 mm
 Primary: Ir_rms 10.354 A => Minimum trace width = 7.96 mm
 Secondary: Io_dc 25.0 A => Minimum trace width = 26.87 mm
 Secondary: Ios_rms 30.545 A => Minimum trace width = 35.42 mm

Layer position = outer , copper = 1 oz (Thickness = 34 μ m)
 Primary: Iin_dc 4.286 A => Minimum trace width = 1.18 mm
 Primary: Ir_rms 10.354 A => Minimum trace width = 3.98 mm
 Secondary: Io_dc 25.0 A => Minimum trace width = 13.43 mm
 Secondary: Ios_rms 30.545 A => Minimum trace width = 17.71 mm

Layer position = inner , copper = 0.5 oz (Thickness = 17 μ m)
 Primary: Iin_dc 4.286 A => Minimum trace width = 6.14 mm
 Primary: Ir_rms 10.354 A => Minimum trace width = 20.72 mm
 Secondary: Io_dc 25.0 A => Minimum trace width = 69.89 mm
 Secondary: Ios_rms 30.545 A => Minimum trace width = 92.13 mm

Layer position = inner , copper = 1 oz (Thickness = 34 μ m)
 Primary: Iin_dc 4.286 A => Minimum trace width = 3.07 mm
 Primary: Ir_rms 10.354 A => Minimum trace width = 10.36 mm
 Secondary: Io_dc 25.0 A => Minimum trace width = 34.95 mm
 Secondary: Ios_rms 30.545 A => Minimum trace width = 46.07 mm

If we use 2 oz in inner and 1 oz in outer layers:

Layer position = outer , copper = 1 oz (Thickness = 34 μ m)
 Primary: Iin_dc 4.286 A => Minimum trace width = 1.18 mm
 Primary: Ir_rms 10.354 A => Minimum trace width = 3.98 mm
 Secondary: Io_dc 25.0 A => Minimum trace width = 13.43 mm
 Secondary: Ios_rms 30.545 A => Minimum trace width = 17.71 mm

Layer position = inner , copper = 2 oz (Thickness = 68 μ m)
 Primary: Iin_dc 4.286 A => Minimum trace width = 1.53 mm
 Primary: Ir_rms 10.354 A => Minimum trace width = 5.18 mm
 Secondary: Io_dc 25.0 A => Minimum trace width = 17.47 mm
 Secondary: Ios_rms 30.545 A => Minimum trace width = 23.03 mm

Gerber file:	gerbers_v52_Y6	Build Time:	3-4 days
Base Material:	FR-4	Layers:	4
Dimension:	140 mm* 300 mm	PCB Qty:	5
Product Type:	Industrial/Consumer electronics	Different Design:	1
Delivery Format:	Single PCB	PCB Thickness:	1.6mm
Specify Stackup:	no No requirement	Layer Sequence:	
PCB Color:	Green	Silkscreen:	White
Material Type:	TG135	Via Covering:	Plugged
Surface Finish:	HASL(with lead)	Deburring/Edge rounding:	No
Outer Copper Weight:	1 oz	Inner Copper Weight:	2 oz
Gold Fingers:	No	Electrical Test:	Flying Probe Fully Test
Castellated Holes:	no	Press-Fit Hole:	No
Edge Plating:	No	Mark on PCB	Order Number
Blind Slot:	No	Min via hole size/diameter:	0.3mm/(0.4/0.45mm)
4-Wire Kelvin Test:	No	Paper between PCBs:	No
Appearance Quality:	IPC Class 2 Standard	Confirm Production file:	No
Silkscreen Technology:	Ink-jet Printing Silkscreen	Package Box:	With JLCPCB logo
Inspection Report:	No	Board Outline Tolerance:	±0.2mm(Regular)
UL Marking:	No	Countersink Hole:	No
Backdrill:	No	Shortage Preference:	Require Full Quantity

Figure 2 JLC-configuration of this PCB

Example of trace widths in the actual layout

In the actual design, the PCB is FR4 1.6mm with 4 layers:

- 2 oz of copper in the inner layers
- 1 oz of copper in the outer layers

Below some check of the minimum regions widths :

Primary side

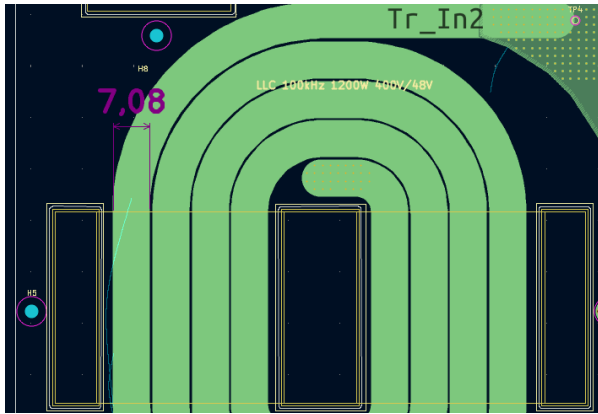


Figure 3 One inner layer

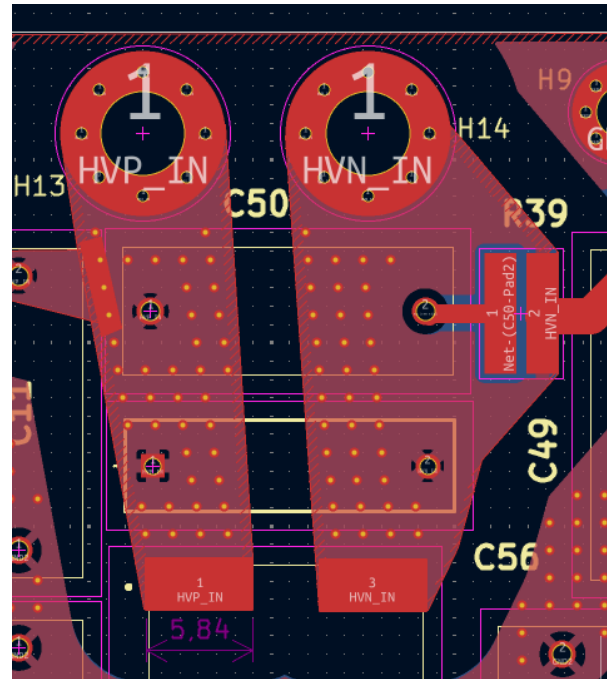


Figure 4 2 outer layers

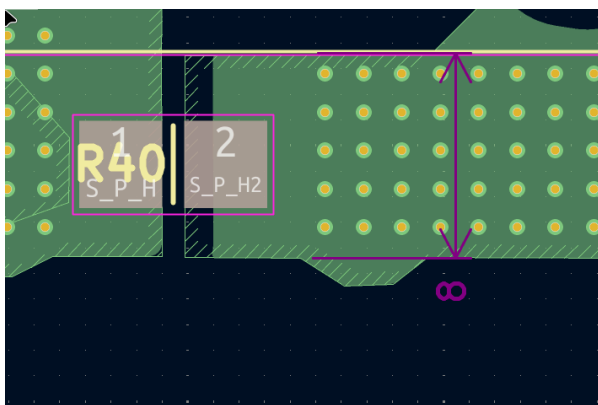


Figure 5 4 layers

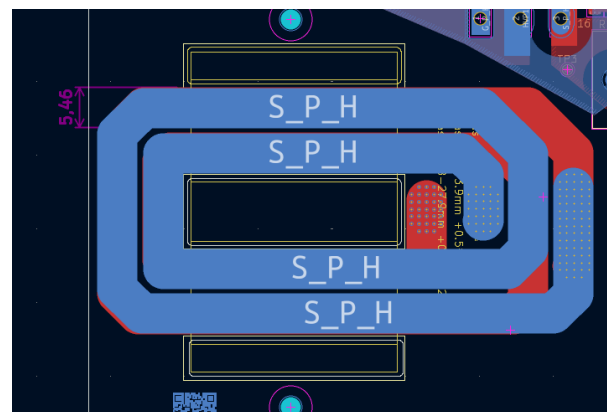


Figure 6 One layer by trace

All trace widths are OK.

Secondary side

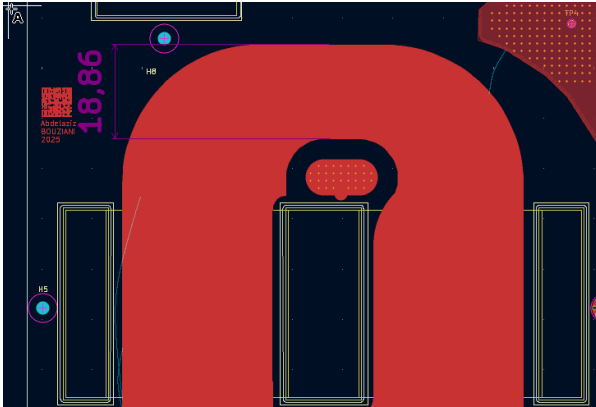


Figure 7 One outer layer

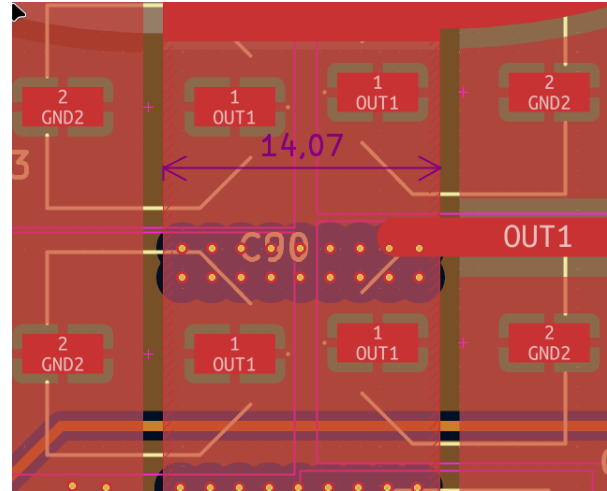


Figure 8 3 layers

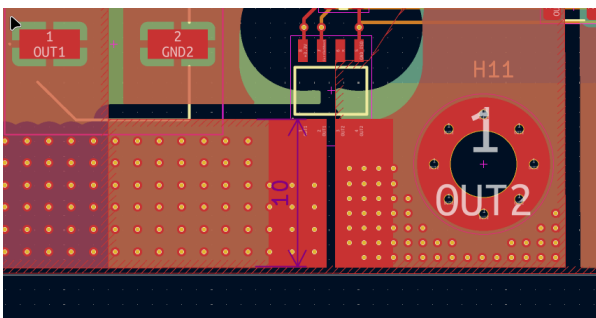


Figure 9 4 layers

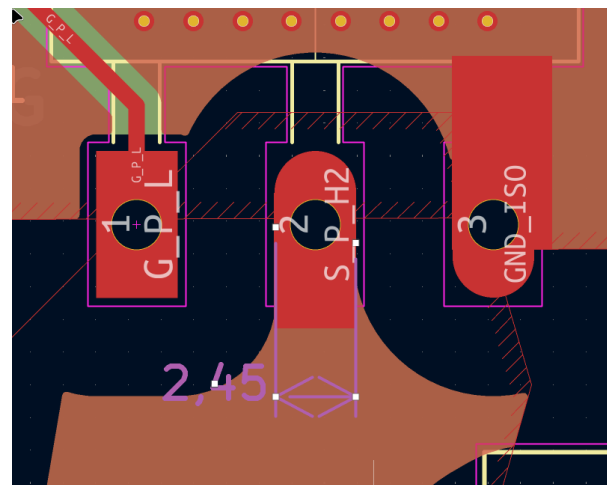


Figure 10 4 layers

Insulation distance

Rules

Conductor spacing (Differential voltage)

Design rules : below the table of conductor spacing, see IPC-2221A

Table 6-1 Electrical Conductor Spacing

Voltage Between Conductors (DC or AC Peaks)	Minimum Spacing						
	Bare Board				Assembly		
	B1	B2	B3	B4	A5	A6	A7
0-15	0.05 mm [0.00197 in]	0.1 mm [0.0039 in]	0.1 mm [0.0039 in]	0.05 mm [0.00197 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]
16-30	0.05 mm [0.00197 in]	0.1 mm [0.0039 in]	0.1 mm [0.0039 in]	0.05 mm [0.00197 in]	0.13 mm [0.00512 in]	0.25 mm [0.00984 in]	0.13 mm [0.00512 in]
31-50	0.1 mm [0.0039 in]	0.6 mm [0.024 in]	0.6 mm [0.024 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.4 mm [0.016 in]	0.13 mm [0.00512 in]
51-100	0.1 mm [0.0039 in]	0.6 mm [0.024 in]	1.5 mm [0.0591 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.5 mm [0.020 in]	0.13 mm [0.00512 in]
101-150	0.2 mm [0.0079 in]	0.6 mm [0.024 in]	3.2 mm [0.126 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
151-170	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	3.2 mm [0.126 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
171-250	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	6.4 mm [0.252 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
251-300	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	12.5 mm [0.4921 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.8 mm [0.031 in]
301-500	0.25 mm [0.00984 in]	2.5 mm [0.0984 in]	12.5 mm [0.4921 in]	0.8 mm [0.031 in]	0.8 mm [0.031 in]	1.5 mm [0.0591 in]	0.8 mm [0.031 in]
> 500 See para. 6.3 for calc.	0.0025 mm /volt	0.005 mm /volt	0.025 mm /volt	0.00305 mm /volt	0.00305 mm /volt	0.00305 mm /volt	0.00305 mm /volt

B1 - Internal Conductors

B2 - External Conductors, uncoated, sea level to 3050 m [10,007 feet]

B3 - External Conductors, uncoated, over 3050 m [10,007 feet]

B4 - External Conductors, with permanent polymer coating (any elevation)

A5 - External Conductors, with conformal coating over assembly (any elevation)

A6 - External Component lead/termination, uncoated, sea level to 3050 m [10,007 feet]

A7 - External Component lead termination, with conformal coating (any elevation)

Figure 11

Hypothesis:

- Differential voltage = 600V (400V with margin)
- B2 : External conductors, uncoated, over 3050m

$$\text{Spacing}_{\text{Norm}} = 0.005 \text{ (mm/V)}$$

$$\text{Voltage} = 600 \text{ (V)}$$

$$\text{Spacing} = \text{Spacing}_{\text{Norm}} \cdot \text{Voltage} = 0.005 \cdot 600 = 3.000 \text{ (mm)}$$

Common mode voltage (safety)

Hypothesis:

- Common mode (transitory peak voltage) = 5kV
- Classe A, pollution grad = 3mm

Tableau F.2 — Distances d'isolement pour supporter les surtensions transitoires

Tension de tenue aux chocs prescrite ^{1) 5)}	Distances minimales d'isolement dans l'air jusqu'à 2 000 m au-dessus du niveau de la mer					
	Cas A Champ hétérogène (voir 3.15)			Cas B Champ homogène (voir 3.14)		
	Degré de pollution ⁶⁾			Degré de pollution ⁶⁾		
	1 mm	2 mm	3 mm	1 mm	2 mm	3 mm
kV						
0,33 ²⁾	0,01	0,2 ^{3) 4)}	0,8 ⁴⁾	0,01	0,2 ^{3) 4)}	0,8 ⁴⁾
0,40	0,02			0,02		
0,50 ²⁾	0,04			0,04		
0,60	0,06			0,06		
0,80 ²⁾	0,10			0,10		
1,0	0,15			0,15		
1,2	0,25	0,25		0,2		
1,5 ²⁾	0,5	0,5		0,3	0,3	
2,0	1,0	1,0	1,0	0,45	0,45	
2,5 ²⁾	1,5	1,5	1,5	0,60	0,60	
3,0	2,0	2,0	2,0	0,80	0,80	
4,0 ²⁾	3,0	3,0	3,0	1,2	1,2	1,2
5,0	4,0	4,0	4,0	1,5	1,5	1,5
6,0 ²⁾	5,5	5,5	5,5	2,0	2,0	2,0
8,0 ²⁾	8,0	8,0	8,0	3,0	3,0	3,0
10	11	11	11	3,5	3,5	3,5
12 ²⁾	14	14	14	4,5	4,5	4,5
15	18	18	18	5,5	5,5	5,5
20	25	25	25	8,0	8,0	8,0
25	33	33	33	10	10	10
30	40	40	40	12,5	12,5	12,5
40	60	60	60	17	17	17
50	75	75	75	22	22	22

Figure 12

Insulation distance = 4mm Bellow the table of insulation between

Application to the actual design

gnd 48v hvn hvm hvp

gnd	0.5	NaN	NaN	NaN	NaN
48v	1.0	1.0	NaN	NaN	NaN
hvn	4.0	4.0	0.5	NaN	NaN
hvm	4.0	4.0	3.0	2.0	NaN
hvp	4.0	4.0	3.0	3.0	0.5

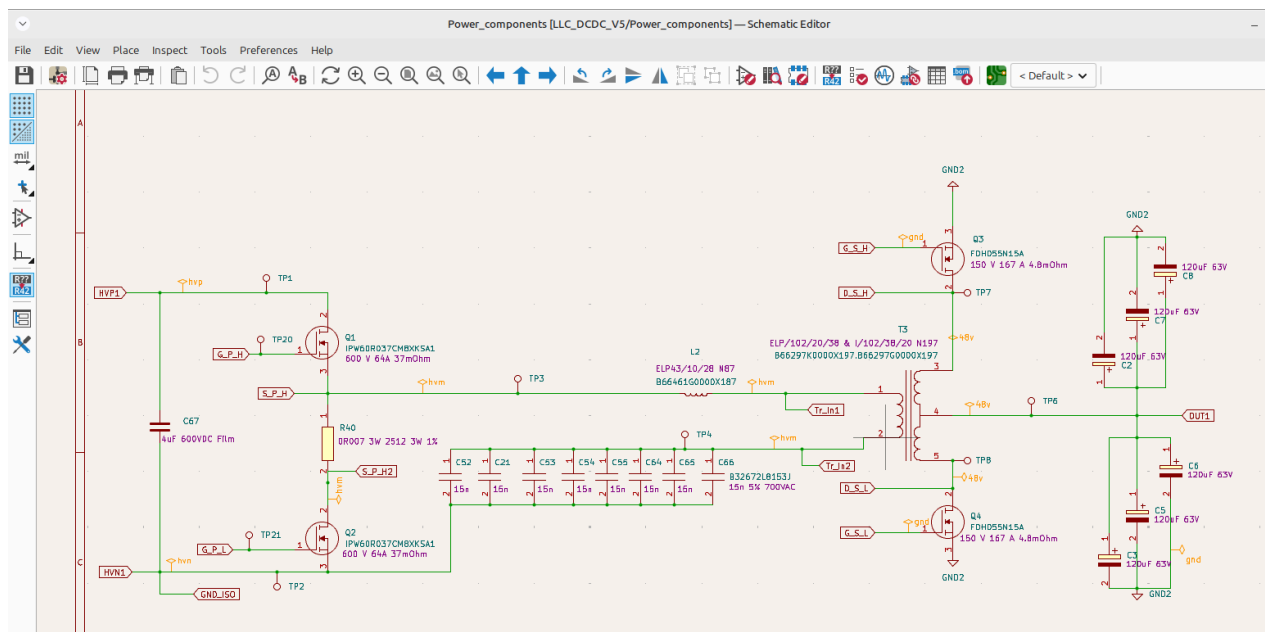


Figure 13 Example of the power zoon of the schematic: 'Net Class' are in Orange

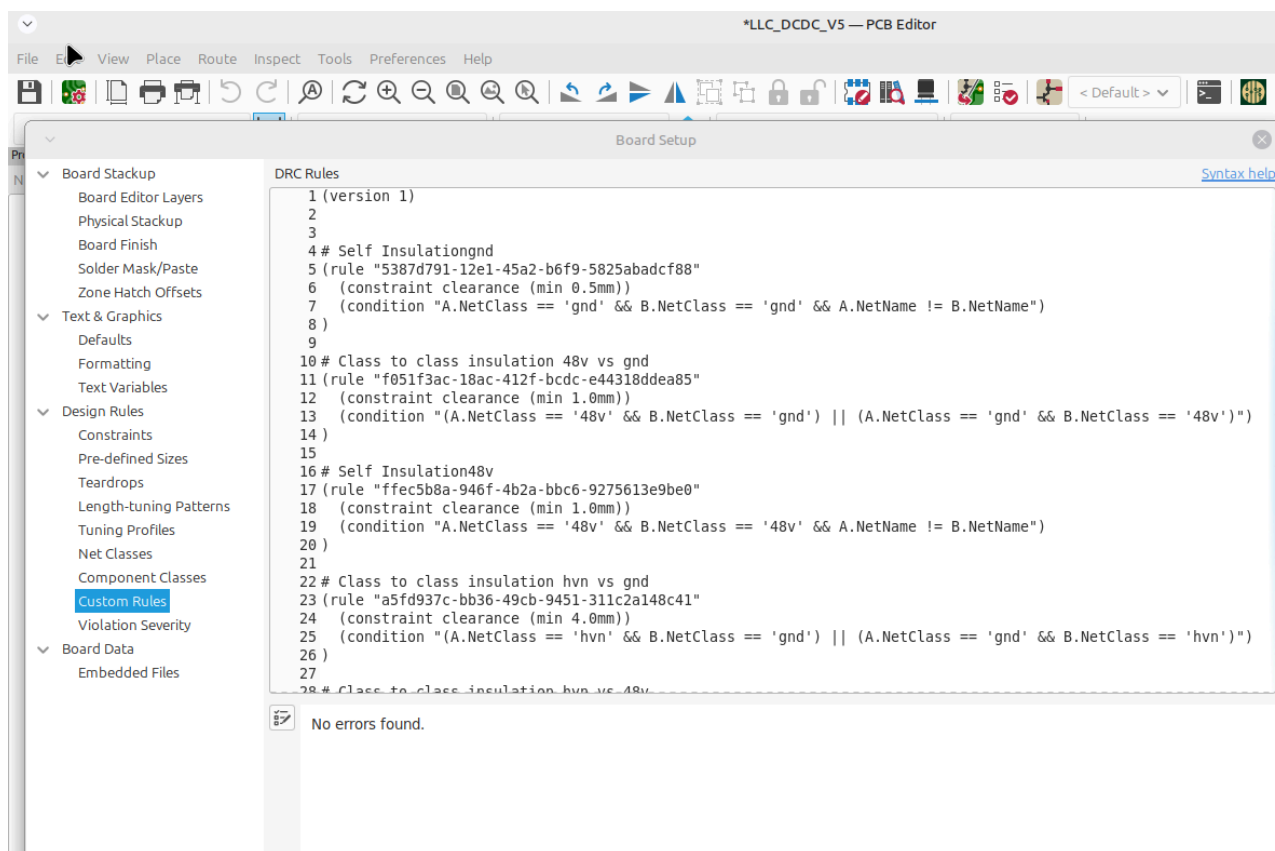


Figure 14 KiCad PCB: isolation rules